Sigflow: an automated and comprehensive pipeline for cancer genome mutational signature analysis

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Abstract: Mutational signatures reflect the accumulated effects of both exogenous and endogenous mutational processes acting on cancer cells. Mutational signature analysis has been becoming a routine procedure after somatic variant calling in cancer genome study. Here we present Sigflow pipeline, which provides an one-stop solution for de novo signature extraction, reference signature fitting, signature stability analysis, and etc. in different types of genome DNA alterations.

Related publications:

– Wang, S., Tao, Z., Wu, T., & Liu, X. S. (2020). Sigflow: an automated and comprehensive pipeline for cancer genome mutational signature analysis. bioRxiv.

– Wang, S., Li, H., Song, M., He, Z., Wu, T., Wang, X., ... & Liu, X. S. (2020). Copy number signature analyses in prostate cancer reveal distinct etiologies and clinical outcomes. medRxiv.

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Previous work, challenge, and approach

Mutational signatures have been initially identified by Alexandrov and colleagues with nonnegative matrix factorization (NMF) based matrix decomposition algorithm in 2013. Different other types of algorithms such as Bayesian NMF, Expectation–maximization have also been built for do novo mutational signature extraction. The application of mutational signature analysis to ever-growing amount of sequencing data leads to the formation of COSMIC signature database.

Current available mutational signature analysis tools either provide limited analysis features, or only focus on specific type of genome alterations, such as single base substitutions (e.g. MutationalPatterns).

Besides, absolute signature exposure (a.k.a. signature contribution) is one of the most important measures in signature analysis, but most of tools can only output relative exposure quantification and visualization (e.g. deconstructSigs).

In addition, the installation and application processes of currently available tools are complex and time-consuming.

Techniques and Methods

Sigflow is highly customizable with many parameter settings and is well supported for different input file formats, and all options are explained in the integrated help section or use cases. It has been designed to run as a command line based program with a user-friendly interface, which allows non-expert users to become quickly familiarized.

To solve the complex and time-consuming installation issues accompanying with current bioinformatics tools, a Docker image of Sigflow is constructed, and this enables good scalability for addition of other analysis features or other types of signatures in the future.

Sigflow begins with importing somatic variant data in MAF (recommended), VCF or CSV/EXCEL format, and then parses the user input to select the workflow to run. Subsequently, a sample by mutation catalogue matrix is generated. Finally, a user specified workflow is performed to extract and analyze mutational signatures. Important immediate and final results are saved to disk for general use. In addition, parallel computation is enabled, and this significantly reduces computation time.

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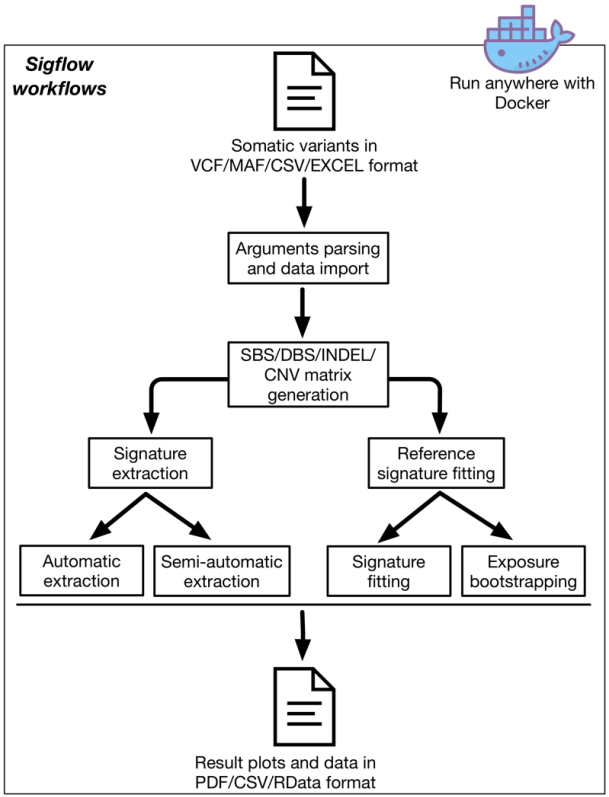
Results and Conclusions

A comprehensive mutational signature analysis pipeline tool Sigflow:

- Run environment: Windows/MacOS/Linux/Docker
- Run way: command line interface
- GitHub Repository: https://github.com/ShixiangWang/sigflow
- Co-evolutionary R package: sigminer
- Major features:
  - *De novo* SBS、DBS、ID and copy number signature extraction
    - Semi-automatic way
      - Standard NMF
    - Automatic ways
      - Bayesian NMF
      - SigProfiler caller
  - All or specified COSMIC reference signatures fitting
  - Signature stability analysis
  - Signature exposure based sample clustering
  - Absolute and relative signature exposure quantification
  - Signature and exposure profile visualization



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